

Intelligent Complaint Classification Using Localized Transformer Architectures (E-Commerce): Data Card

Section	Your response
Input vs. Output	Input: Natural-language customer complaints and customer-support queries in text form. Output: A predicted complaint category for each input text. The system focuses on complaint classification and routing rather than complaint resolution.
Classification problem	This is a multi-class text classification task. The model classifies each complaint into one of 10 categories: billing, product_defect, delivery_shipping, refund_return, customer_service, account_access, fraud_unauthorized, warranty_repair, subscription_cancel, and general_inquiry. These categories cover common e-commerce issues involving payments, products, delivery, refunds, accounts, fraud, warranties, subscriptions, and general support.
Labeling	Labels are derived from the existing labels in the source datasets and mapped into the project's common taxonomy. The Bitext customer-support intents are mapped to the target categories, while CFPB fraud complaints provide examples for fraud_unauthorized. Consistency and accuracy are checked through stratified data splitting and evaluation using precision, recall, F1-score, macro F1, and per-class performance. Particular attention is given to less-represented classes and class imbalance.
Data creation	The dataset is created by combining existing, publicly available labeled complaint and customer-support datasets rather than generating synthetic complaints. The sources include the Bitext Customer Support Training Dataset, CFPB consumer complaint data, and the provided labeled complaint dataset. The datasets are standardized into a common text-label structure and split using stratified sampling.
Web collection ethics	The project uses publicly available datasets, including Bitext customer-support data and CFPB consumer complaint data. The current approach uses existing datasets rather than directly scraping websites. No AI-generated data or crowd workers are used to create the dataset. The project statement identifies publicly available complaint data as an alternative to accessing private company data. The original materials do not establish the consent and licensing conditions for every source, so these should be documented before redistribution. Personal information should also be appropriately anonymized.
Bias and representation	The dataset may underrepresent consumers whose language, writing style, communication channel, or complaint type is poorly represented in the

	source datasets. Some categories may also have fewer examples than others. Multiple datasets are combined to broaden coverage, while stratified sampling preserves class representation across splits and class weighting reduces the influence of imbalanced classes. Macro and per-class F1 scores will be used to identify poor performance on minority categories. This aligns with the project's commitment to fairness across both common and less-represented complaint categories.
Values and community alignment	The dataset supports the project's values of accessibility, efficiency, fairness, customer satisfaction, transparency, accountability, and responsible AI . Natural-language classification improves accessibility and can reduce manual triage, while class balancing and per-class evaluation support fairness. Reporting precision, recall, and F1-score supports transparency. Human oversight is retained for uncertain, sensitive, or high-impact complaints, supporting accountability. The expected community benefit is faster and more consistent complaint routing and reduced workload for customer-support personnel.